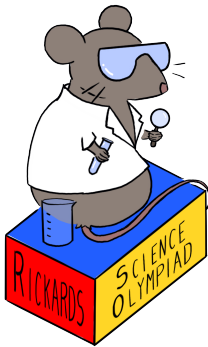


# Codebusters C

## Rickards Invitational

1-8 November, 2025



This test was sponsored by the FSU Computer Science Department and College of Arts and Sciences

- You will have **50 minutes** to complete this test.
- You may bring up to **(3)** Class I or II calculators, as well as any writing utensils.
- You may use the space on the exam as scratch paper, but please make sure that your answers are clear. Either write your answers separately and box them, or write them directly over/under the ciphertext.
- You may take apart the test as you wish, however you should write your team number **on top of every page** and a **250 point deduction** will be taken if the pages are **not** put back in the correct order.
- The first question on the exam is **Timed**. If you believe you have solved the question in the first 10 minutes, signal to the event supervisor who will come over to check it. If you are correct, your time will be recorded and a bonus issued; if you have more than 2 errors you will be notified and allowed to retry for a bonus throughout the first 10 minutes.
- If you successfully complete the timed question after 10 minutes, you may still get the base points for the question however it will not be checked and no bonus will be awarded.
- There are 3 questions marked for **Special Bonus**: #17, #23, #32

Team Name: \_\_\_\_\_

Team Number: \_\_\_\_\_

Written By: \_\_\_\_\_

Note: All listed accomplishments are in Codebusters specifically

**Aneesh Iyer** (GaTech '28, National Silver Medalist), **Claire Lin** (WWPN '26, National Bronze Medalist), **Anam Ali** (LKSOM, PA States Champion), **Arthur Armaing** (GaTech '28, National t10), **Chinmay Govind** (Penn '28, PA States Champion), **Eddy Zhao** (Cal '29, MIT Medalist), **Emir Naduvil** (CMU '29, 2x National 7th), **Jeffrey Xu** (Rutgers '28, 2024 National Champion), **Roha Herial** (Penn '29, 2025 National Champion), **Samuel Wang** (Stony '29, 2023 National Champion), **Sean Chen** (GaTech '29, NorCal State Champion)

## Scoring Table

### Exam Score

| Question | Points | Score |
|----------|--------|-------|
| 1        | 363    |       |
| 2        | 282    |       |
| 3        | 238    |       |
| 4        | 259    |       |
| 5        | 223    |       |
| 6        | 473    |       |
| 7        | 274    |       |
| 8        | 253    |       |
| 9        | 211    |       |
| 10       | 256    |       |
| 11       | 274    |       |

| Question | Points | Score |
|----------|--------|-------|
| 12       | 519    |       |
| 13       | 264    |       |
| 14       | 251    |       |
| 15       | 486    |       |
| 16       | 294    |       |
| 17       | 308    |       |
| 18       | 332    |       |
| 19       | 216    |       |
| 20       | 268    |       |
| 21       | 229    |       |
| 22       | 344    |       |

| Question | Points | Score |
|----------|--------|-------|
| 23       | 368    |       |
| 24       | 202    |       |
| 25       | 248    |       |
| 26       | 267    |       |
| 27       | 267    |       |
| 28       | 458    |       |
| 29       | 332    |       |
| 30       | 244    |       |
| 31       | 361    |       |
| 32       | 293    |       |
| Total:   | 9657   |       |

### Timed Question

Question Score: Circle one: 250      0

Time: \_\_\_\_\_  
minutes : seconds

Timed Bonus:  $(600 - \text{time in seconds}) \times 2 = \text{timed bonus}$

Total TQ Score:  $\text{question score} + \text{timed bonus} = \text{total TQ score}$

### Special Bonus

Circle one. Special bonus is only awarded for solving the question for full score. Special bonus questions: 17, 23, 32.

- 1 Correct = 150
- 2 Correct = 400
- 3 Correct = 750
- 0 Correct = 0

### Total

Final Score =  $\text{Exam Score Total} + \text{TQ Total} + \text{Special Bonus} =$  \_\_\_\_\_

Rank: \_\_\_\_\_

## Tables

The tables below may be useful.

|       |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Keys  | a | b | c | d | e | f | g | h | i | j | k | l | m |
| ----- |   |   |   |   |   |   |   |   |   |   |   |   |   |
| A,B   | n | o | p | q | r | s | t | u | v | w | x | y | z |
| C,D   | o | p | q | r | s | t | u | v | w | x | y | z | n |
| E,F   | p | q | r | s | t | u | v | w | x | y | z | n | o |
| G,H   | q | r | s | t | u | v | w | x | y | z | n | o | p |
| I,J   | r | s | t | u | v | w | x | y | z | n | o | p | q |
| K,L   | s | t | u | v | w | x | y | z | n | o | p | q | r |
| M,N   | t | u | v | w | x | y | z | n | o | p | q | r | s |
| O,P   | u | v | w | x | y | z | n | o | p | q | r | s | t |
| Q,R   | v | w | x | y | z | n | o | p | q | r | s | t | u |
| S,T   | w | x | y | z | n | o | p | q | r | s | t | u | v |
| U,V   | x | y | z | n | o | p | q | r | s | t | u | v | w |
| W,X   | y | z | n | o | p | q | r | s | t | u | v | w | x |
| Y,Z   | z | n | o | p | q | r | s | t | u | v | w | x | y |

|     |       |
|-----|-------|
| A   | AAAAA |
| B   | AAAAB |
| C   | AAABA |
| D   | AAABB |
| E   | AABAA |
| F   | AABAB |
| G   | AABBA |
| H   | AABBB |
| I/J | ABAAA |
| K   | ABAAB |
| L   | ABABA |
| M   | ABABB |

|     |        |
|-----|--------|
| N   | ABBAA  |
| O   | ABBAB  |
| P   | ABBB A |
| Q   | ABBBB  |
| R   | BAAAA  |
| S   | BAAAB  |
| T   | BAABA  |
| U/V | BAABB  |
| W   | BABAA  |
| X   | BABAB  |
| Y   | BABBA  |
| Z   | BABBB  |

|   |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
|---|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|----|
| A | B | C | D | E | F | G | H | I | J | K  | L  | M  | N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| Z | Y | X | W | V | U | T | S | R | Q | P | O | N | M | L | K | J | I | H | G | F | E | D | C | B | A |

|   |   |    |    |   |    |    |    |    |    |    |    |
|---|---|----|----|---|----|----|----|----|----|----|----|
| 1 | 3 | 5  | 7  | 9 | 11 | 15 | 17 | 19 | 21 | 23 | 25 |
| 1 | 9 | 21 | 15 | 3 | 19 | 7  | 23 | 11 | 5  | 17 | 25 |

## Frequency of English Letters

|            |           |           |           |           |
|------------|-----------|-----------|-----------|-----------|
| E - 12.51% | S - 6.54% | C - 3.06% | G - 1.96% | K - 0.67% |
| T - 9.25%  | R - 6.12% | U - 2.71% | W - 1.92% | X - 0.19% |
| A - 8.04%  | H - 5.49% | M - 2.53% | Y - 1.73% | J - 0.16% |
| O - 7.60%  | L - 4.14% | F - 2.30% | B - 1.54% | Q - 0.11% |
| I - 7.26%  | D - 3.99% | P - 2.00% | V - 0.99% | Z - 0.09% |
| N - 7.09%  |           |           |           |           |

## Frequency of Spanish Letters

|            |           |           |           |           |
|------------|-----------|-----------|-----------|-----------|
| E - 14.08% | I - 5.98% | M - 3.08% | Y - 1.09% | Z - 0.47% |
| A - 12.16% | L - 5.24% | P - 2.89% | V - 1.05% | Ñ - 0.17% |
| O - 9.20%  | U - 4.69% | B - 1.49% | G - 1.00% | X - 0.14% |
| S - 7.20%  | D - 4.67% | H - 1.18% | F - 0.69% | K - 0.11% |
| N - 6.83%  | T - 4.60% | Q - 1.11% | J - 0.52% | W - 0.04% |
| R - 6.41%  | C - 3.87% |           |           |           |

For the purposes of cryptograms it is customary to treat n and ñ as distinct letters, but a and á are the same letter. Likewise for e and é, and i and í. In other words, all the accent marks get amputated when working with cryptograms. Also, while some older Spanish dictionaries consider ch, ll, and rr, to be their own letters—this has fallen out of modern usage. Accordingly, “burro” is considered as five letters: “b-u-r-r-o” and not as four letters “b-u-rr-o.”

## Morse Code

|   |      |
|---|------|
| A | •—   |
| B | —••• |
| C | —•—• |
| D | —••  |
| E | •    |

|   |      |
|---|------|
| F | ••—• |
| G | —•—• |
| H | •••• |
| I | ••   |
| J | •—•— |

|   |      |
|---|------|
| K | —•—  |
| L | •—•• |
| M | —•—  |
| N | —•   |
| O | —•—• |

|   |      |
|---|------|
| P | •—•— |
| Q | —•—• |
| R | •—•  |
| S | •••• |
| T | —    |

|   |      |
|---|------|
| U | ••—  |
| V | •••— |
| W | •—•— |
| X | —••— |
| Y | —•—• |
| Z | —•—• |

|   |       |
|---|-------|
| 0 | —•—•— |
| 1 | •—•—• |

|   |       |
|---|-------|
| 2 | ••—•— |
| 3 | •••—• |

|   |       |
|---|-------|
| 4 | ••••— |
| 5 | ••••• |

|   |       |
|---|-------|
| 6 | —•••• |
| 7 | —•—•• |

|   |       |
|---|-------|
| 8 | —•—•• |
| 9 | —•—•• |

|       |   |
|-------|---|
| •     | E |
| ••    | I |
| •••   | S |
| ••••  | H |
| —•••  | B |
| —•—•  | 0 |
| ••••• | 5 |

|       |   |
|-------|---|
| —     | T |
| •—    | A |
| ••—   | U |
| •••—  | V |
| —••—  | X |
| •—•—  | 1 |
| —•••• | 6 |

|       |   |
|-------|---|
| —•    | N |
| •—•   | R |
| ••—•  | F |
| —•—•  | C |
| ••—•  | 2 |
| —•••• | 7 |

|       |   |
|-------|---|
| —•    | M |
| •—•   | W |
| ••—•  | L |
| —•—•  | Y |
| •••—  | 3 |
| —•••• | 8 |

|      |   |
|------|---|
| —••  | D |
| •—•  | P |
| —••  | Z |
| •••— | 4 |
| —••• | 9 |

|     |   |
|-----|---|
| —•— | K |
| •—• | J |
| —•— | Q |

|     |   |
|-----|---|
| —•— | G |
| —•— | O |

**Timed Question.** [250 Points] Solve this **Aristocrat**. When you have finished, **watch [this Youtube Video](#)** so your answer can be checked and time recorded.

K MHU YSHY DXTI JHBW UHM ZHPPKCJ RW, PKOW H RHJCWY

YX RWYHP, VWRHCVKCJ YSHY UW RXNW YXJWYSWI, MPXUPD.

|             | A | B | C | D | E | F | G | H  | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W  | X | Y | Z |
|-------------|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|
| Frequency   | 0 | 1 | 4 | 2 | 0 | 0 | 0 | 10 | 2 | 5 | 4 | 0 | 3 | 1 | 1 | 6 | 0 | 5 | 3 | 1 | 4 | 2 | 10 | 5 | 9 | 1 |
| Replacement |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |

1. (363 points) Solve this **K2 Xenocrypt** about triumph.

WY GOJUU JU MYUZJ, WY JHQHRJ JR VQJGY. ZYWRJ GJ

ZQJ, ZYCÑOJ BPYCB FYVQJWAB OWB DBRBUUB JZQFB.

|             |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|-------------|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| Replacement |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| K2          | A | B | C | D | E | F | G | H | I | J  | K | L | M | N | Ñ | O | P | Q | R | S | T | U | V | W | X | Y | Z |  |
| Frequency   | 1 | 8 | 2 | 1 | 0 | 2 | 3 | 1 | 0 | 14 | 0 | 1 | 1 | 0 | 1 | 3 | 1 | 5 | 4 | 0 | 0 | 5 | 2 | 5 | 0 | 8 | 5 |  |

2. (282 points) Solve this **K2 Aristocrat Cypher** that has some spelling irregularities.

AJERSCDLWEU J VRRDSL JLRCEQ BVS JEESGS RT BVS

BVWLBH-NBRLSH NYHNMLJOSL BRRY JSREN BR AJNBSL.

|             |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |   |   |  |  |
|-------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|----|---|---|---|---|---|---|---|--|--|
| Replacement |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |    |   |   |   |   |   |   |   |  |  |
| K2          | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R  | S  | T | U | V | W | X | Y | Z |  |  |
| Frequency   | 2 | 8 | 2 | 2 | 6 | 0 | 1 | 3 | 0 | 7 | 0 | 8 | 1 | 5 | 1 | 0 | 1 | 10 | 10 | 1 | 1 | 4 | 2 | 0 | 2 | 0 |  |  |

3. (238 points) Solve this **Aristocrat** about betrayal.

SRSJ VM PWEUS KCASJX, UEVSEJS NZGN A NCYUNSX, EJS

QZE UZGCSX VM DCSGX, ZGU NYCJSX GIGAJUN VS.

|             | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S  | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|
| Frequency   | 3 | 0 | 5 | 1 | 5 | 0 | 6 | 0 | 1 | 6 | 1 | 0 | 2 | 6 | 0 | 1 | 1 | 1 | 12 | 0 | 6 | 4 | 1 | 5 | 2 | 4 |
| Replacement |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |

4. (259 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **ETIDI**.

EDCIO TALLS DRAXT SBQDT RDYOI EBXHU EUNOE WDAET

ATOEM TDKHR NEIAE X

5. (223 points) Decode this phrase that was encoded using the **Porta** cipher with a keyword of **SEVEN**.

JASLA QTMWX NNXLW GCXJY FCJWX OXLTO EDSIV QPDRX

FUIFV YTID

6. (473 points) Solve this **K1 Patristocrat** about taking the first step.

AWUIB KUIAM JJFWC VIIOK AAWUI WVIDR WRMAP CJJNW

VPKOI VNIKV FWCBR IVUMF HIJKP IDKJJ HIIMA KIZ

| K1          | A | B | C | D | E | F | G | H | I  | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 6 | 2 | 3 | 2 | 0 | 3 | 0 | 2 | 14 | 7 | 7 | 0 | 4 | 2 | 2 | 3 | 0 | 3 | 0 | 0 | 4 | 6 | 7 | 0 | 0 | 1 |
| Replacement |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |



7. (274 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **HEARTBEAT**.

$$\begin{pmatrix} H & E & A \\ R & T & B \\ E & A & T \end{pmatrix} = \begin{pmatrix} 7 & 4 & 0 \\ 17 & 19 & 1 \\ 4 & 0 & 19 \end{pmatrix} \quad \begin{pmatrix} 7 & 4 & 0 \\ 17 & 19 & 1 \\ 4 & 0 & 19 \end{pmatrix}^{-1} = \begin{pmatrix} 25 & 18 & 10 \\ 15 & 1 & 15 \\ 18 & 14 & 13 \end{pmatrix}$$

YVUWAOGNYWUWQLIVDYOWMCGHDVGIFN

- 
8. (253 points) Decode this phrase that was encoded using the **Porta** cipher. You are told the plaintext ends with **LLY**.

VKHDX VTBBY XXJQX GJAQG NPRWW CODAV RVVKH UWEXK

BDTOQ H

9. (211 points) Decode this phrase that was encoded using the **Baconian** cipher.

ABYZABYAZBAYZBYZABABYAZYBABZAYBAZBABAYBABZAYBZABAYBAZBA

BYAZYBZYZAYBAZBYAZYBABABZYABZYZABYAZBAYBABZYZAYZBYZABAY

BAZBABYABABZAYBAZYBZYABZYABZABYAZBABYABABZAYBAZBYABZABY

AZYBABZYAZYBZABAYBZYABZAB

10. (256 points) Solve this **Aristocrat** about having patience.

QW IPLOIQI LEVIOMBSI GEO KFJMNNEFBLQIBL FJ QMLSTIK

EBVW YW LTI KIMVIO'J FBMVFVFLW LE ZFDI QI GMSI

SMOKJ.

|             | A | B | C | D | E | F | G | H | I  | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 0 | 5 | 0 | 1 | 5 | 7 | 2 | 0 | 13 | 4 | 4 | 8 | 7 | 2 | 5 | 1 | 5 | 0 | 4 | 2 | 0 | 4 | 4 | 0 | 2 | 1 |
| Replacement |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

11. (274 points) Solve this **K2 Aristocrat**. You are told that the keyword used is 6 letters long. What is the keyword?

**Box** your final answer.

YZ XGKKLA OZC RXGWW EZS XGE MLLW, EZSA HLPYN PR

PAALQWGILGHWL, G QALRLYIL KOGK IGYYZK HL IZQPLJ ZA

RSHRKPSKLJ.

|             |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |
|-------------|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|--|--|
| Replacement |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |  |
| K2          | A | B | C | D | E | F | G | H | I | J | K | L  | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |  |  |  |
| Frequency   | 6 | 0 | 1 | 0 | 3 | 0 | 8 | 4 | 4 | 2 | 8 | 13 | 1 | 1 | 2 | 5 | 3 | 5 | 4 | 0 | 0 | 0 | 5 | 3 | 5 | 7 |  |  |  |

12. (519 points) Solve this **K2 Patristocrat** about an average introvert.

HXTER TPCAE JTFKP GGRPF TSEUH EEHXS ERHEF GPCSS

PFVCT GGQIH CEHXY CWHTF FYUYT SXYNA YVTHP AVYCW

HEKEN TC

|             |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Replacement |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| K2          | A | B | C | D | E  | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| Frequency   | 3 | 0 | 7 | 0 | 10 | 6 | 5 | 9 | 1 | 1 | 2 | 0 | 0 | 2 | 0 | 6 | 1 | 3 | 5 | 9 | 2 | 3 | 2 | 4 | 6 | 0 |

13. (264 points) Decode this phrase that was encoded using the **Nihilist Substitution** cipher with a keyword of **ENGLISH** and a polybius key of **TEACHER**.

43 59 36 48 62 88 57      24 56 69 74 66 75 59

33 66 39 48 42 59 27      47 46 40 45 52 56 46

56 70 66 47 46 57 50      26 47 56 58 72 66 36      53 67 37

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 |   |   |   |   |   |
| 2 |   |   |   |   |   |
| 3 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 5 |   |   |   |   |   |

14. (251 points) Solve this **cryptarithm** for 902154 635. Write out your final answer and  it. ★ *This question is a special bonus question.*

Base 10 Addition

Answer: 902154 635 (two words)

```

    DISTANCE
+  ISOLATION
-----
    DISTASTES

```

15. (486 points) Decode this phrase that was encoded using the **Nihilist Substitution** cipher. You are told that the keyword used to encode it is between 3 and 7 letters long and the plaintext ends with **YOU'RE ALIVE**.

37 23 70 86 68 64 66 46 78 66 24 44 58 86 75 48

34 57 66 45 65 54 58 86 76 36 53 60 55 49 65 45

60 68 56 47 34 58 56 46 28 64 49 68 57 35 47 88

68

|   | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| 1 |   |   |   |   |   |
| 2 |   |   |   |   |   |
| 3 |   |   |   |   |   |
| 4 |   |   |   |   |   |
| 5 |   |   |   |   |   |

16. (294 points) Solve this **K1 Aristocrat**. You are told that the keyword used is 5 letters long. What is the keyword?

your final answer.

SXTPNCPO FSCI KDAKVP BPADW, CIP MJXJXJ NAPJCDAPB

MPNYWP QAPXHSPQ JXQ SXOPBCADNCS MVP.

| K1          | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P  | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 6 | 3 | 6 | 4 | 0 | 1 | 0 | 1 | 2 | 5 | 2 | 0 | 3 | 4 | 4 | 13 | 1 | 0 | 5 | 1 | 0 | 2 | 2 | 6 | 1 | 0 |
| Replacement |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |

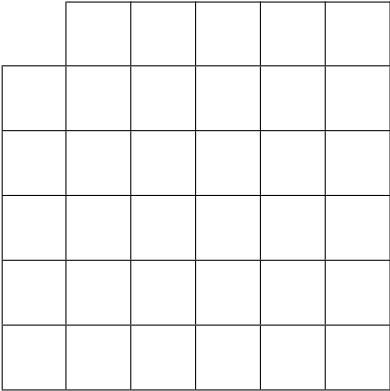
17. (308 points) Decode this phrase that was encoded using the **Checkerboard** cipher. You are told that the plaintext ends with **EACH OTHER**. ★*This question is a special bonus question.*

DY AN DW RW RD RY OW DD AD AY RI AI AN DY RW RY

AN RD AY RY RD RY RD DD RD SD DW SW AY AI OW AY

RN OY OW DD RD RN AW DW DD RD OW OY AN DW DY AN

RD DD



18. (332 points) Solve this **K3 Aristocrat**. You are told that the keyword used is 8 letters long. What is the keyword?  

Box

 your final answer.

TQB’P RUP MQIW TWUXAF SUP PQQ OXW XVUXT QO MQI.

FPXMGBS GB PVU CWUFUBP GF XF GACQWPXBP OQW FIZZUFF

XF MQIW SQXRF.

|             |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |
|-------------|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|
| K3          | A | B | C | D | E | F  | G | H | I | J | K | L | M | N | O | P | Q  | R | S | T | U | V | W | X | Y | Z |
| Frequency   | 2 | 5 | 2 | 0 | 0 | 10 | 4 | 0 | 4 | 0 | 0 | 0 | 4 | 0 | 3 | 9 | 10 | 2 | 3 | 3 | 8 | 2 | 7 | 9 | 0 | 2 |
| Replacement |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |

19. (216 points) Solve this **Aristocrat** about a mesmerizing gaze.

AQJLAFA HAOCJP VOCUAU JBA SBITA CZKXALUA JI UJIY

OZN UJOLA GIL O SBKTA, HCJ IZTP IZA SKTT HA JBA

TCVEP FOZ.

|             | A  | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 14 | 4 | 5 | 0 | 1 | 2 | 1 | 3 | 6 | 8 | 3 | 4 | 0 | 1 | 6 | 3 | 1 | 0 | 3 | 6 | 5 | 2 | 0 | 1 | 1 | 5 |
| Replacement |    |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

20. (268 points) Solve this **Aristocrat** about a tragedy.

UGYMQZHA YFKN OHVJV JGUUAM TGHVMV GZ UPUMQ-JTFK

BGSMQ XPQNV PKN VMKN MXG-ZQFMKNAD UAPVJFX RPOV

NQFZJFKO.

|             | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 4 | 1 | 0 | 1 | 0 | 6 | 6 | 3 | 0 | 5 | 6 | 0 | 8 | 6 | 3 | 5 | 6 | 1 | 1 | 2 | 6 | 8 | 0 | 3 | 2 | 4 |
| Replacement |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

21. (229 points) Solve this **Aristocrat** about limitations.

GUMMHNY NG MBCWNUDIVNP VT UMTMUKMH GNU CMNCWM

SVIJNFI GDYVWVDW HFIVMT JNWHVPE IJMY ODRX.

|             | A | B | C | D | E | F | G | H | I | J | K | L | M  | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 0 | 1 | 3 | 4 | 1 | 2 | 4 | 4 | 5 | 3 | 1 | 0 | 10 | 8 | 1 | 2 | 0 | 1 | 1 | 3 | 5 | 7 | 5 | 1 | 3 | 0 |
| Replacement |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |   |

22. (344 points) Solve this **K1 Xenocrypt** that was taken straight from the soul.

WÑ XLJÑQ QK XL PIHÑEIM KNJÑ KM KW MIXOHK QK ZMÑ

PLZQÑQ YZK KN WÑ PÑGLJÑW QK ZMÑ MÑPLIM LMNZWÑH.

| K1          | A | B | C | D | E | F | G | H | I | J | K | L | M | N | Ñ  | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 3 | 4 | 3 | 9 | 6 | 8 | 3 | 12 | 1 | 4 | 6 | 0 | 0 | 0 | 0 | 0 | 5 | 3 | 1 | 5 |
| Replacement |   |   |   |   |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |   |



23. (368 points) Solve this **cryptarithm** for 307154 2968. Write out your final answer and  it. ★ *This question is a special bonus question.*

Base 10 Complex Number Multiplication

Answer: 307154 2968 (two words)

$$\begin{aligned}
 &(WA - Ti)(E + Ri) \\
 &= WA \cdot E + (WA \cdot R)i - (T \cdot E)i + T \cdot R \\
 &= LYE + LRYi - WHi + CR \\
 &= CHI + LLYi
 \end{aligned}$$

24. (202 points) Solve this **Aristocrat** about a legendary night.

IF EYMG LDEJLCB FMHYE, ICX NDEEBJ OMBB FIE TJDGJ

CFEMB OJ TDETY D HBMQZGJ IL EYJ RDOF'G XDVG.

|             | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
|-------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Frequency   | 0 | 6 | 3 | 7 | 9 | 5 | 5 | 2 | 4 | 7 | 0 | 3 | 5 | 1 | 3 | 0 | 1 | 1 | 0 | 3 | 0 | 1 | 0 | 2 | 4 | 1 |
| Replacement |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

25. (248 points) Decode this phrase that was encoded using the **Hill** cipher and the encoding key **TOUR**.

$$\begin{pmatrix} T & O \\ U & R \end{pmatrix} = \begin{pmatrix} 19 & 14 \\ 20 & 17 \end{pmatrix}$$

QTYCLPWRENUXQCHYSQSAWVMCJY

- 
26. (267 points) Decode this phrase that was encoded using the **Complete Columnar** cipher. You are told that the plaintext contains **OTHRO**.

GRTUI OIBGA EAOHE HEIOP REHDT EYAHH GOASW DAURO

NHHRN GNRST NGTTI O

27. (267 points) Decode this phrase that was encoded using the **Checkerboard** cipher with a polybius keyword of **SATURDAY**.

IO EE TE TY IY      TO IM ME TO EO      EE EN IM IN TO

IN IO TO EN IM      SN IY EM EM MN      TO SO IY EM EE

TN TM EE MM MM      TE EN TN EE EN      ME MO MN TN

|  |  |  |  |  |  |
|--|--|--|--|--|--|
|  |  |  |  |  |  |
|  |  |  |  |  |  |
|  |  |  |  |  |  |
|  |  |  |  |  |  |
|  |  |  |  |  |  |
|  |  |  |  |  |  |

28. (458 points) Solve this **K2 Patristocrat**. You are told that the plaintext contains the word **DIABETES**.

SJMOF IROMO VSJNJ LVRES EIEBO ODDSJ DESOB JEKID

SDSOT ZERXT ZJZND SONTJ KODOE JDDSO EJYOD TYO

|             |   |   |   |    |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |
|-------------|---|---|---|----|---|---|---|---|---|---|---|---|---|---|----|---|---|---|---|---|---|---|---|---|---|---|
| Replacement |   |   |   |    |   |   |   |   |   |   |   |   |   |   |    |   |   |   |   |   |   |   |   |   |   |   |
| K2          | A | B | C | D  | E | F | G | H | I | J | K | L | M | N | O  | P | Q | R | S | T | U | V | W | X | Y | Z |
| Frequency   | 0 | 2 | 0 | 10 | 8 | 1 | 0 | 0 | 3 | 9 | 2 | 1 | 2 | 3 | 13 | 0 | 0 | 3 | 9 | 4 | 0 | 2 | 0 | 1 | 2 | 3 |



